Project Design Phase-I
Solution Architecture

Date	29 October 2026
Team ID	N/A - individual project; no Team ID provided
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

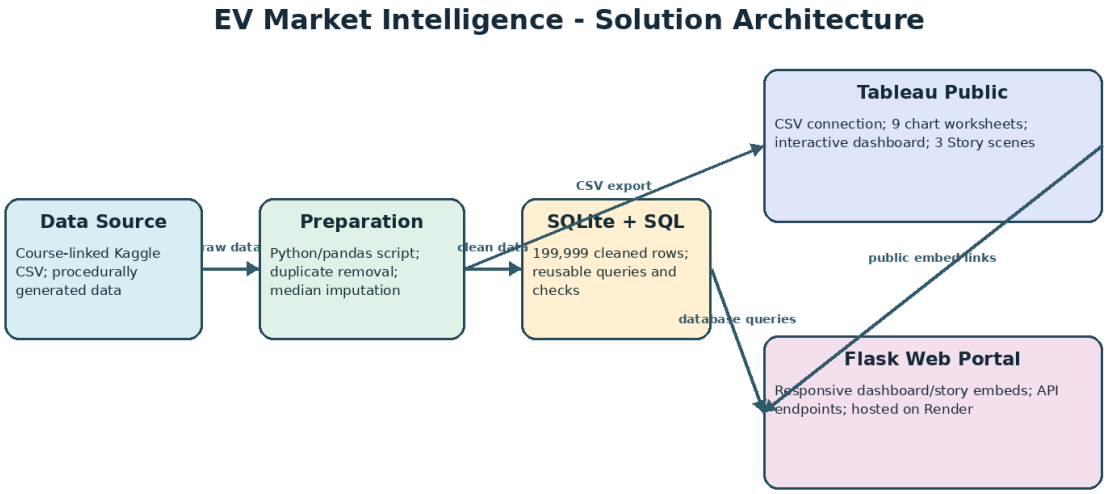
- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Sample Solution Architecture Diagram:

Component Description Table

Component Name	Description / Role in Architecture	Technology Used

EV Market Intelligence architecture (project-specific)



Component Description Table

Component	Technology / Artifact	Responsibility
Dataset source	Kaggle source supplied by SkillWallet; synthetic CSV dataset	Provides vehicle, manufacturer, region, battery and pricing attributes.
Preparation	Python/pandas script	Removes one duplicate row and imputes median values for missing numeric cells.
SQLite database	data/ev_market.db; ev_market_clean	Stores 199,999 cleaned rows and supports grouped SQL queries and quality checks.
Tableau workbook	CSV connection in Tableau Public	Nine visualizations; dashboard filters; three Story scenes. Not a direct live SQLite connector.
Flask app	flask_app/app.py and templates/index.html	Renders the homepage and embeds published Tableau views; supplies KPI/region/battery endpoints.
Hosting	Render	Runs the Flask application using Gunicorn, with a health check.
Repository	GitHub	Stores source, requirements, configuration example and documentation; confirm public repository link before final hand-in.